Haebeom **Jung**

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Education

Seoul National University — *Seoul, South Korea*

Ph.D. Student in Artificial Intelligence — *Sep. 2023 - Present*

- **Current GPA**: 4.04/4.30
- A member of the SNU VGI Lab (Advisor: Prof. Jaesik Park)
- Received full scholarship for 5 semesters.

KwangWoon University — *Seoul, South Korea*

B.S. in Division of Robotics — *Mar. 2016 - Feb. 2023*

- **GPA**: 3.81/4.50, **Major GPA**: 3.93/4.50
- Received full scholarship for 6 semesters.

Publications

Beyond the Current Scene: Event-Referential Grasping with Active View Selection — *arXiv 2026*

Hyunjoon Lee*, **Haebeom Jung***, Eunsung Cha, Daeun Lee, Yu-Chiang Frank Wang, Jaesung Choe, and Jaesik Park — *Sep. 2026*

(* equal contribution)

Distractor-Aware COLMAP-Free 3D Gaussian Splatting — *Under Review*

Namtae Kim, **Haebeom Jung**, and Jaesik Park

Articulated Object Reconstruction from Rest-State Observation — *ECCV 2026*

Daeun Lee, Jaeah Lee, Woosung Kim, **Haebeom Jung**, and Jaesik Park — *Sep. 2026*

Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting — *RA-L 2026*

Haebeom Jung, Namtae Kim, Jungwoo Kim, and Jaesik Park — *Apr. 2026*

Metropolis-Hastings Sampling for 3D Gaussian Reconstruction — *NeurIPS 2025*

Hyunjin Kim, **Haebeom Jung**, and Jaesik Park — *Dec. 2025*

Professional Experience

Kakao Mobility — *Pangyo, South Korea*

Software · Robotics Engineer (Full-time) — *Dec. 2021 - Apr. 2022*

- Developed a data collection system for autonomous vehicles.
- Researched visual SLAM and localization for real-world driving environments.

Stryx — *Seoul, South Korea*

Software · Robotics Engineer (Compulsory Military Service) — *Dec. 2019 - Dec. 2021*

- Led software development for the hardware business unit.
- Developed a sensor data collection system for a mobile mapping platform.
- Researched and developed autonomous navigation for unmanned ground vehicles, including SLAM, localization, sensor fusion, and path planning.

ROBOTIS — *Seoul, South Korea*

Robotics Engineer (Intern) — *Jan. 2019 - Feb. 2019*

- Validated ROBOTIS OpenManipulator prior to its open-source release.
- Developed and published an example application for ROBOTIS OpenManipulator SARA.

Skills

Programming	Python, C/C++, Java
Robotics & Tools	PyTorch, ROS/ROS 2, Isaac Sim, Qt
Languages	Korean (native), English (fluent)

Open Source Contributions

Selected Robotics Projects

- Author of **TLC-Calib**, the official implementation of Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting (RA-L 2026).
- Maintainer of **ORB-SLAM3-ROS2**, a ROS 2 integration for the ORB-SLAM3 library.
- Contributor to **FAST-LIO**, fixing Velodyne LiDAR integration issues.
- Contributor to **LIO-SAM**, adding a ROS service for map saving.

Patents

Device and Method for Targetless LiDAR-Camera Calibration — *Seoul National Univ., South Korea*

Haebeom Jung, Namtae Kim, and Jaesik Park — *May. 2025*

Application No. 10-2025-0067792

Method of estimating the location of a moving object using vector map — *Kakao Mobility, South Korea*

Yesung Hwang, **Haebeom Jung**, Jeongmin Kang, Ilsuk Park, Seungkwan Song, and Seunghwan Hong — *Jan. 2024*

Registration No. 10-2624-6440000

Awards

Robotics Competitions

International

2018	**3rd Place**, ROBOCUP CANADA, KidSize Humanoid Technical Challenge **[Video]**	*Montreal, Canada*
2017	**4th Place**, ROBOCUP IRAN, KidSize Humanoid Soccer **[Video]**	*Tehran, Iran*

Domestic

2018	**3rd Place**, R-BIZ Challenge, Turtlebot3 Autorace **[Video]**	*Ilsan, South Korea*
2018	**2nd Place**, IRC, Intelligence Humanoid Robot Sports	*Ilsan, South Korea*
2017	**Winner**, IRC, Intelligence Humanoid Robot Sports **[Video]**	*Ilsan, South Korea*
2017	**Winner**, ROBOCUP KOREA, Intelligence Humanoid Soccer	*Daegu, South Korea*
2016	**3rd Award**, KIRO, Future-Robot Idea Presentation	*Pohang, South Korea*

Honors

2019	**Outstanding Mentor**, 2019 Samsung Electronics Junior Software Creation Contest	*Samsung Electronics*
2017	**Achievement Award**, Spread the name of KwangWoon University throughout the year	*KwangWoon Univ.*

Scholarships

2026	**Full scholarship**, Jeongheon Foundation Scholar	*Jeongheon Foundation*
2023-2025	**Full scholarship**, AI Fellowship	*Seoul National Univ.*
2017-2019	**Full scholarship**, Robotics Competition Team	*KwangWoon Univ.*

Fellowships and Research Funding

Photo-Realistic Rendering for Surrounding Dynamic Mobile Scenes — *Ministry of Education, South Korea*

Master's Research Fellowship Program — *Jul. 2024 - Jun. 2025*

Artificial Intelligence Graduate School Program — *Ministry of Science and ICT, South Korea*

Seoul National University, IPAI — *Sep. 2023 - Jul. 2025*

Research Service & Robotics Activities

Invited Talks

- NAVER LABS, Apr. 2025

Reviewer

- CVPR (2026)
- ECCV (2026)
- NeurIPS (2026)
- IEEE TIP (2025)

Kakao Mobility NEMO Conference

Seoul, South Korea

Autonomous driving robot docent

Feb. 2022

- Explained and operated an autonomous robot that travels inside the conference hall. **[Video]**

Samsung Electronics Junior Software Creation Contest

Suwon, South Korea

Mentor

Jul. 2019 - Sep. 2019

- Serviced the app, which provided a system for managing kindergarten students going to and from school and attendance using beacons.
- The high school team I coached won the grand prize among 10,000 teams.

RO:BIT (Robot Game Sports Team)

Seoul, South Korea

Member & Team Leader at 2018

Jul. 2016 - Dec. 2018

- As a member of the humanoid team for 3 years, I participated in many external activities and competitions related to robots.
- I mainly researched and developed a robot algorithm called humanoid intelligence in games such as soccer, basketball and athletics, etc.
- In 2018, I led the team as the leader of the humanoid team and won awards in international competitions.